

Pediatric Pneumonia Classification from Chest X-rays

Kermany CXR · Guangzhou Women and Children's Medical Center

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LITERATURE SURVEY · BASELINE · HYPOTHESIS · 2018–2026

INTRODUCTION

What these slides cover.

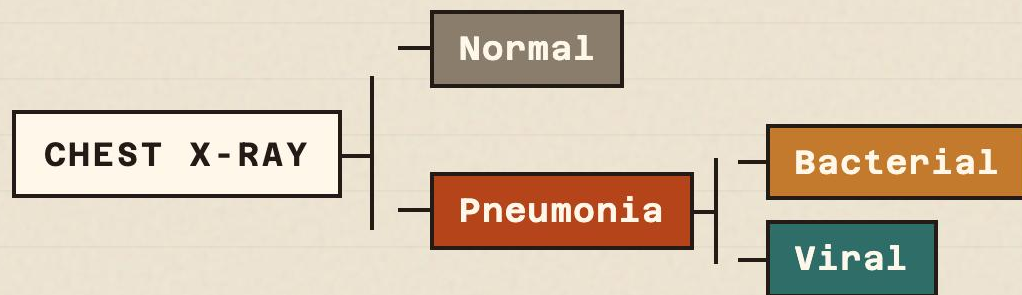
01 The data. One chest X-ray is normal or pneumonia. Pneumonia splits into bacterial and viral. The films are from Guangzhou. The children are 1 to 5 years old.

02 Ten papers, 2018 to 2026. Each paper uses its own split. These slides do not rank the papers on one accuracy number.

03 Our baseline. We reproduce XCCNet on the full Kermany set. The last slide states one hypothesis and one prediction. We state both before the next code change.

THE DATASET

Three labels on one film.



5,232

TRAIN IMAGES

624

TEST PATIENTS

1–5 yr

PEDIATRIC CXRS

Guangzhou

WOMEN & CHILDREN'S MEDICAL
CENTER

Literature findings

01 Kermany et al., 2018 · *Cell*

02 Rajaraman et al., 2018 · *Appl. Sci.*

03 Stephen et al., 2019 · *J. Healthc. Eng.*

04 Rahman et al., 2020 · arXiv / *Appl. Sci.*

05 Salehi et al., 2021 · *Br. J. Radiol.*

06 Ayan, Karabulut & Ünver, 2021 · *Arab. J. Sci. Eng.*

07 Kundu et al., 2021 · *PLoS One*

08 XCCNet, 2024 · arXiv

09 Chauhan, Gupta & Doja, 2025 · LightPneumoNet

10 Transfer learning vs custom CNNs, 2026 · arXiv

Kermany et al.

INCEPTION V3 FROZEN

2 BINARIES

OFFICIAL 624



MODEL / TASK

Inception V3, ImageNet, conv frozen, softmax only. Pneumonia vs normal + bacterial vs viral. No 3-class.

PREPROCESS

Quality screening. Remove unreadable images.

IMBALANCE

None. Live with 3:1 and 2:1.

TASK	ACC	SENS	SPEC	AUC
P vs N	92.8%	93.2%	90.1%	96.8%
B vs V	90.7%	88.6%	90.9%	94.0%

Rajaraman et al.

CUSTOM VGG16

CUT LAST LAYER · GAP + DENSE

2 BINARIES + 3-CLASS

SAME 624



MODEL / TASK

Best: customized VGG16 (ImageNet). Cut the last layer, add GAP + dense. Also scratch sequential / residual / Inception. Both binaries + 3-class. CAM / LIME.

PREPROCESS

Atlas lung crop. Full film + ROI → 1024², mean-norm. No geo aug.

IMBALANCE

None. Keep noisy train images.

TASK (VGG16, CROP)	ACC	AUC	RECALL	F1 / SPEC
P vs N	96.2%	0.993	96.2%	F1 0.970
B vs V	93.6%	0.962	98.4%	Spec 86.0%
3-class	91.8%	0.939	90.0%	F1 0.910

Stephen et al.

SCRATCH CNN 1 BINARY ONLY RESHUFFLED SPLIT



MODEL / TASK	PREPROCESS	IMBALANCE
4-block CNN from scratch. Sigmoid. Pneumonia vs normal only. Best input 200×200×3.	Resize, 1/255. Rot 40°, shift/shear/zoom 0.2, h-flip. No lung crop.	Heavy Keras aug. No class weights.

SPLIT	ACC	LOSS	AUC / SENS
Train	95.31%	0.129	—
Validation	93.73%	0.184	not reported

Rahman et al.

DENSENET201 WINS

2 BINARIES + 3-CLASS

OWN ~200/CLASS TEST



MODEL / TASK	PREPROCESS	IMBALANCE		
AlexNet, ResNet18, DenseNet201, SqueezeNet. MATLAB. All three tasks. 5-fold.	227 ² or 224 ² . ImageNet norm. Rot 45°, 10% scale/shift. No crop.	Augment every class to 4,500.		
DENSENET201	ACC	RECALL	AUC	F1
P vs N	98.0%	99%	0.98	0.981
B vs V	95.0%	96%	0.952	0.952
3-class	93.3%	93.2%	0.95	0.935

Salehi et al.

DENSENET121

BINARY ONLY

HONEST / LOWER

■ N ■ P · SAME KERMANY POOL, ~3:1



MODEL / TASK

VGG19, DenseNet121, Xception, ResNet50. Best DenseNet121. Pneumonia vs normal only.

PREPROCESS

1/255, rot 15°, small shift/shear/zoom, flips, denoise, contrast, mean subtract.

IMBALANCE

SMOTE (rare in this field).

MODEL	ACC	AUC	SENS
DenseNet121	86.8%	0.86	>91%
Xception	86.0%	0.81	>91%
All four	≥83%	—	>91%

Ayan, Karabulut & Ünver

ENSEMBLE OF 3 BINARY + 3-CLASS ~624 TEST



MODEL / TASK
VGG / ResNet / Inception / Xception / MobileNet / SqueezeNet / PNet. Best = 3 models, not 7. Binary + 3-class.

PREPROCESS
TL, GAP, batch-norm, L2. Rot / zoom / flip on normals.

IMBALANCE
Normals 1,349 → 2,534 (near bacterial count).

TASK	ACC	SENS / REC	SPEC / F1
Binary ensemble	95.83%	97.76%	Spec 92.73% · AUC 95.21%
3-class ensemble	90.71%	B 97.93%	—
Viral class	—	77.03%	F1 82.31%

Kundu et al.

3-CNN ENSEMBLE

BINARY ONLY

KERMANY + RSNA



MODEL / TASK

PREPROCESS

IMBALANCE

Weighted-prob ensemble: GoogLeNet + ResNet-18 + DenseNet-121. Pneumonia vs normal.

Resize 224×224×3. No SMOTE.

Transfer learning only. No reweight.

SET	ACC	PREC	RECALL	F1 / AUC
Kermamy	98.81%	98.82%	98.80%	F1 98.79% · AUC 98.35%
RSNA	86.95%	—	87.02%	drops ~12 pts

XCCNet

CNN + TRANSFORMER

BINARY

80/10/10 NOT 624



MODEL / TASK

PREPROCESS

IMBALANCE

Dilated CNN (SFx) + contrastive transformer (CoTFx). Binary. Grad-CAM / Score-CAM / LIME.

CLAHE, lung seg (ResUNet++), rib suppression.

GAN/ADA: +2,624 synthetic normals (1,583 → 4,207).

SETTING	ACC	P / R / F1
No aug	96.29%	—
Traditional aug	97.18%	—
Full stack	99.76%	99.75%

LightPneumoNet · Chauhan et al.

388K-PARAM CNN

BINARY

INDEPENDENT TEST

■ N ■ P · WEIGHTS 2.0 / 1.2



MODEL / TASKScratch 4-block CNN, filters 16→128, 1.48 MB. Pneumonia vs normal.

PREPROCESS224², RGB→gray, [0,1]. Rot 12°, zoom/shift/shear 0.15. No h-flip.

IMBALANCEclass_weight Normal 2.0, Pneumonia 1.2.

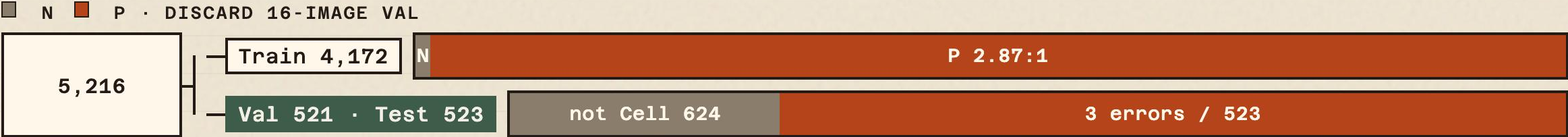
TASK	ACC	PREC	RECALL	F1
P vs N	94.2%	0.92	0.99	0.96

TL vs custom CNNs

RESNET50 FINE-TUNE

BINARY

80/10/10 OF 5,216



MODEL / TASK

Scratch 4-block vs ResNet50 / DenseNet121 / EfficientNet-B0. Frozen vs last-2-block FT. Binary.

PREPROCESS

224², gray→3ch, ImageNet mean/std. Flip, ±10° rot, affine, jitter.

IMBALANCE

None. Report the 2.87:1 ratio.

RESNET50 FT	ACC	F1	AUC	SENS / SPEC
Headline	99.43%	99.61%	99.93%	99.48% / 99.26%
Note	FT beats frozen +5.48% avg. Ensemble added 0. Different exam than 624.			

DO NOT RANK ON ONE BOARD

Same set. Different exams.

#	PAPER	TASK	IMBALANCE	HEADLINE	SPLIT
1	Kermany	2 bin	none	92.8 / 90.7	624
2	Rajaraman	2 bin + 3	noisy train	96.2 / 93.6 / 91.8	624
3	Stephen	1 bin	heavy aug	93.7 val	custom
4	Rahman	2 bin + 3	4,500/cls	98 / 95 / 93.3	~200/cls
5	Salehi	1 bin	SMOTE	86.8	conservative
6	Ayan	bin + 3	aug N	95.8 / 90.7 · viral 77	~624
7	Kundu	1 bin	TL	98.8 → 87 RSNA	2 sets
8	XCCNet	1 bin	GAN	99.76	80/10/10
9	LightPneumo	1 bin	weights	94.2 / rec 0.99	held-out
10	2026 TL	1 bin	none	99.43	reshuffle

Hypothesis

We train XCCNet on all 5,856 Kermany films. The split is stratified 80/10/10. The run uses CLAHE and a lung mask. CLIP stays frozen. This run does not include the GAN. This run does not include rib suppression.

RESULT	ACC	PREC	RECALL	F1
Our test, 585 films	94.87%	94.81%	98.36%	96.55%
Paper, traditional aug	97.18%	94.52%	98.65%	96.54%
Paper, full stack	99.76%	99.75%	99.75%	99.75%

The model misses 7 pneumonia films and marks 23 normal films as pneumonia. The 99.76% test adds 2,624 synthetic normal films, so the two classes are almost equal.

Hypothesis. The gap comes from the GAN and from rib suppression. **Prediction.** Add one of those two stages on this same split. Accuracy on the 585-film test rises above 94.87%. The same model on the official 624-patient test scores below 94.87%.